

Department of Mathematics and Statistics

MTH 343

Assignment 1 - Fall 2024

The homework must be submitted to ybelhamadia@aus.edu

By Sunday September 29th 2024 at 23 :59.

1. Müller's method is a root finding method that uses three initial approximations, p_0 , p_1 , and p_2 , and determines the next approximation p_3 by considering the intersection of the x -axis with the parabola through $(p_0, f(p_0))$, $(p_1, f(p_1))$, and $(p_2, f(p_2))$. The algorithm considers three given initial approximations p_0 , p_1 and p_2 , and generate the value of p_3 based on the following

$$p_3 = p_2 - \frac{2c}{b + \operatorname{sgn}(b)\sqrt{b^2 - 4ac}},$$

where

$$c = f(p_2),$$

$$b = \frac{(p_0 - p_2)^2 [f(p_1) - f(p_2)] - (p_1 - p_2)^2 [f(p_0) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)},$$

$$a = \frac{(p_1 - p_2) [f(p_0) - f(p_2)] - (p_0 - p_2) [f(p_1) - f(p_2)]}{(p_0 - p_2)(p_1 - p_2)(p_0 - p_1)}.$$

Then continue the iteration, with p_1 , p_2 and p_3 replacing p_0 , p_1 and p_2 .

Write an M-file function that implements the Müller's method for root-finding problem. The input parameters should be : the function f , three initial guesses, an error tolerance, and the maximum number of iterations. The algorithm should stop when the approximate relative error is less than or equal to the tolerance. Your output should be a table where each row contains the number of iteration, the value of the root p_n , and the approximate relative error.

2. In order to test your code, compute the root of the equation

$$\cos(x) = 2 \sin(3x)$$

using the initial guesses of $p_0 = 0$, $p_1 = 0.25$ and $p_2 = 0.5$ and with tolerance 10^{-6} , create an M-file script where you can call your program in 1) for this specific example.

Everything should work by simply running your M-file script. The output table should be printed on the screen (to 5 significant figures), and your program should create a plot of the used function and a point representing the root.

3. Your final submission should include :

- (a) **(3 pts)** Your M-file function for question 1)
- (b) **(2 pts)** The main M-file script for question 2). When you run your M-file script, the output table should be printed on the screen, and a plot of the function with the root point should be created.